ECN 593 · APPLIED PROJECT SHORT COURSE · LECTURE 0

Introduction

Prerequisites, notation, and the shape of the argument

ECN 593 short course notes (study draft, not a frozen specification)
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Abstract

These six notes are a short course for one applied project: a network agent-based model is calibrated to World Bank moments by history matching; the image of a named policy contrast over the surviving calibrations is reported as a set; two one-world estimators — a SUTVA learner and an exposure-mapped design-based estimator — are audited against that set. This lecture supplies the prerequisites an economics master's student needs before Lectures 1–5, states the argument from the top down, and freezes the language the rest of the course will not be allowed to abuse. No calibration wave has been run. Nothing here is a result.

1. How to use these notes

Read this lecture first, then 1 through 5 in order. Each later lecture is written to be self-contained on its own objects, but the argument is a composition, and Lecture 5 will not re-teach history matching or Manski. The notes assume first-year graduate training in econometrics — expectations, OLS, potential outcomes as a notation, the idea of an estimand — and almost no training in agent-based models, uncertainty quantification, or partial identification as a research method. Where a term of art is unavoidable it is defined on first use.

The notes are a study draft. They are not a freeze of METHODOLOGY.md, not a proposal to advisors, and not a finding. METHODOLOGY.md remains the specification of the engine. If a sentence here and a sentence there disagree, the specification wins and the notes are wrong. AMENDMENT-01, which adds a second diagnostic estimator, is itself a draft and is labelled as such whenever it appears.

Each lecture ends with a short reading list. You do not need to read the lists before the next lecture. You will want them when a definition here feels thinner than the paper it came from.

2. The project, from the top

The applied project asks a narrow question. A stochastic network model of a preventive behaviour is written down in closed form. It is calibrated not by picking a best-fit parameter vector but by discarding those vectors that cannot reproduce a small set of real cross-country moments, once sampling error, simulation error, and a stated model-discrepancy budget are allowed for. The survivors form a not-ruled-out-yet set in parameter space. A named policy contrast — the change in mean behaviour if one seeds an intervention at network hubs rather than at nobody — is then evaluated at many survivors. The collection of those evaluations is the honest answer. It is a set, not a number.

Separately, one surviving world is treated as if it were the world. A modern heterogeneous-treatment-effect estimator, built under the stable unit treatment value assumption, is fitted to a simulated panel from that world and publishes an interval. A second estimator, built to handle interference and using the known randomisation of

a different assignment rule, publishes another. Each interval is compared with the image of *its own* estimand over the surviving worlds. If a one-world interval sits inside the image and misses part of it, that is a pre-registered finding about overconfidence. If it covers the image, that is a pre-registered finding that the interesting story is false here. If no world survives calibration, that too is a finding, and the protocol forbids loosening the threshold to rescue the exercise.

Two advisors sit on opposite sides of this picture. History matching and model discrepancy are a Durham / computer-experiment tradition. Causal forests and Bayesian causal forests are a causal-machine-learning tradition. The project is interesting, if it is, because it forces those two traditions to share an estimand and then refuses to let either tradition's native interval stand in for the image. The composition is the claim. Equifinality in simulators is not a discovery of this course (Lecture 5 names the near-neighbours). Economist-native policy contrasts under interference, mapped through a history-matched set, with a paired SUTVA foil and an interference-aware foil, on a simulator where the planner's number is known, is the remaining wedge.

3. Potential outcomes, in the form we will use them

For a single unit i and a binary treatment $D_i \in \{0, 1\}$, the Neyman–Rubin notation writes two potential outcomes, $Y_i(1)$ and $Y_i(0)$. The observed outcome is $Y_i = Y_i(D_i)$. The unit-level effect $Y_i(1) - Y_i(0)$ is not observed. Holland (1986) called this the fundamental problem of causal inference. Random assignment of D identifies the average treatment effect $\mathrm{E}[Y_i(1) - Y_i(0)]$ because it makes the two observed conditional means unbiased for the two average potential outcomes. That is the whole trick, and it is enough for most of first-year econometrics.

The trick rests on an assumption that is usually stated once and then ignored. The stable unit treatment value assumption (SUTVA; Rubin 1980, building on Cox 1958) has two parts: there is one version of treatment, and unit i's outcome does not depend on anyone else's treatment. The second part is the one this project is about. If it fails, Y_i is a function of the *entire* assignment vector $\mathbf{D}$,

$$Y_i = Y_i(\mathbf{D}),\ \mathbf{D} \in \{0,1\}^N.$$

There are then 2^{N-1} different "effects of treating i," one for each configuration of the others, and the scalar $Y_i(1) - Y_i(0)$ is not an object of the experiment. Well-posed questions are questions about *assignment rules*: what is the expected population mean under rule π compared with rule π'? Those are the objects Halloran and Struchiner, and later Hudgens and Halloran (2008), organised into direct, indirect, total and overall effects. Lecture 4 returns to the taxonomy. What you need now is only this: once interference is admitted, the estimand must be named as a contrast of policies, not as an average of individual effects.

The project's three rules, in that language

The kernel, written out in METHODOLOGY.md §2, updates each agent's behaviour by a logistic of own lag, a linear-in-means peer term with coefficient γ, covariates, and a treatment shifter that turns on at date zero. The peer term is interference by construction, and $\gamma = 0$ is excluded from the parameter box so that a no-interference world cannot silently survive calibration. Three assignment rules are frozen, all with the same budget of 80 treated agents out of $N = 800$ when anyone is treated:

- π_0, control: $D = 0$ for everyone.
- π_1, random: a simple random sample of 80 agents, without replacement.
- π_2, degree: the 80 highest-degree agents. Deterministic given the graph.

The headline estimand is the overall contrast of hub-seeding against control,

$$\theta_2(c) = E_c[\bar{Y}_T \mid \pi_2] - E_c[\bar{Y}_T \mid \pi_0].$$

$\theta_1(c)$ is the same contrast with π_1 in place of π_2. The targeting premium $\tau(c) = \theta_2(c) - \theta_1(c)$ is zero in any SUTVA world with a constant individual effect, because any 80 treated agents then produce the same mean. A nonzero premium is interference made visible as an estimand. Expectations are over simulation noise only. The rules, given the graph, are deterministic for π_0 and π_2 and are a known design for π_1. That last distinction will matter in Lecture 4: a design-based estimator needs a design, and π_2 does not have one.

4. Identification is not estimation

A parameter, or a functional of a complete-data law, is *point-identified* if the distribution of observables, plus the maintained assumptions, pick out a single value. It is *set-identified* (partially identified) if they pick out a set with more than one point and less than the whole parameter space. It is unidentified if they pick out the whole space. Identification is a statement about the population and the assumptions. It does not mention sample size. Estimation is what one does with a sample once the identified object has been named. A confidence interval is a statement about sampling uncertainty around an identified object. It is not itself an identified set.

Lecture 2 is entirely about that distinction, in Manski's language. What you need now is only the discipline it imposes on later lectures. The image $\Theta_{hm} = \{ \theta_2(c) : c \text{ not ruled out} \}$ will be reported the way one reports an identified set: as the values still compatible with the objects one is willing to maintain. It will not be claimed as a theorem that Θ_{hm} *is* a Manski identified set. The reasons are in Lectures 2 and 5; the short version is that ruling-out uses a thresholded discrepancy, folds sampling and Monte Carlo error into the test, and lives in a complete simulator whose incompleteness is the unknown calibration, not a missing counterfactual.

> **A set is not a distribution**
>
> Putting a uniform (or any) probability on the surviving calibrations, and reporting a mean and standard deviation of $\theta_2(c)$ as if they were posterior moments, is an extra assumption. History matching refuses that assumption. So does partial identification. The notes will not smuggle it back in by calling Θ a "distribution of treatment effects."

5. What the simulator is, and is not

An agent-based model, in the sense this project uses the phrase, is a stochastic recursive map on a finite population. Each of $N = 800$ agents has a state $Y_i(t) \in (0, 1)$, a covariate vector X_i, and a place in an Erdős–Rényi graph drawn once and then frozen. The next state is a logistic of own lag, mean neighbour state, covariates, own treatment, and Gaussian noise. That is the entire "ABM." There are no emergent institutions, no integer programming, no GIS layer, no Mesa schedule. The minimality is intentional. The scientific object is not a realistic society. It is a complete data-generating process in which interference is explicit, cheap to simulate, and impossible for a SUTVA estimator to represent unless neighbour treatment is handed to it as a feature.

Three consequences follow, and they are why a simulator is the right host for the question. First, the planner's number $\theta_{true}(c_0)$ exists, because we can run the kernel at c_0 under both rules as often as we like. In a field study that number is the thing one never sees. Second, the assignment mechanism is known, including when it is

degenerate. Third, “calibration uncertainty” can be separated from “sampling uncertainty in a panel,” because we can hold the world fixed and vary the panel, or hold the panel's design fixed and vary the world. Those two variations are the two legs of the pipeline.

6. Two ways the model is not the world

History matching (Lecture 1) already admits that the kernel, even at the right c, will not reproduce the real system. That admission is the discrepancy budget δ, frozen at two percentage points on each moment. It is an admission about the *simulator*.

A second, logically distinct failure is that the number treated as the outcome is not the thing the estimand named. That is a question about operationalization, not about the kernel. Lecture 3 treats it as an admission restriction on scoring rules: an oracle, a noisy survey, and a bad item that loads on a treatment-channel covariate, of which only the first two pass a pre-declared correlation threshold. The image of the estimand over the admissible scores, at one world, is a second set, Θ_{cv}. It is not a second pass over the not-ruled-out calibrations. Folding the two sets into one “uncertainty interval” would undo the diagnosis.

7. The pipeline, named

Stage	Input	Output	Consumes NROY?
Anchors	World Bank API v2	z (DPT mean 86.2, corr 0.605, internet 68.6)	No
History matching	z, box C, $I(c) \le \tau$	NROY set C	Produces it
Image maps	C, $\theta_2(\cdot)$, $\delta_{spill}(\cdot)$	Θ_{hm}, Θ_{exp}	Yes
Select c_0	C	median-I survivor	As a selection
Simulate panel	c_0, π, kernel	micro-data (Y, D, X)	No
Measurement admission	scores m_1, m_2, m_3	M*, Θ_{cv}	No
Foil 1 (forest / BCF)	panel at c_0	I_{BCF}	No
Foil 2 (exposure map)	π_1 panel at c_0	I_{E2}	No
Oracle	kernel at c_0	$\theta_{true}(c_0)$	No

Table 1. Assembled pipeline. Anchor values are from the 2026-08-12 fetch (scripts/data/anchors.json). Θ_exp and I_E2 exist only if draft AMENDMENT-01 is accepted. No stage after “Anchors” has been run.

The money plot of the project is an overlay of I_{BCF}, I_{E2}, Θ_{hm}, Θ_{exp}, Θ_{cv}, and a tick at $\theta_{true}(c_0)$. Each one-world interval is judged against its own honest set. Cross-plotting the forest interval against Θ_{exp} mixes estimands. Pre-registered verdicts — F-OVERCONF, F-HONEST, F-EMPTY, F-SINGLETON — are defined in METHODOLOGY.md §10 and restated in Lecture 5. All of them are publishable. A paper that can only be written if the forest looks overconfident is not a pre-registered paper.

8. Language discipline

Several compressions were tried, in conversation and in a set of notes produced with an external chatbot, and then rejected. They are listed here so they do not re-enter the later lectures, or the eventual write-up, by habit.

Rejected phrase	Why	Replacement
NROY = observationally valid calibrations	Low I does not certify truth or observational equivalence	Not ruled out yet, given V and τ
I(c) is a stack of moment inequalities	$I \leq \tau$ is one joint ellipsoid, not $E[g] \geq 0$	A pedagogical bridge, not a Pakes–Tamer model
Distribution of treatment effects over NROY	Smuggles a measure onto a set	Image of a named functional; a set
HM assumes perfect measurements	δ exists because they are not	HM assumes the chosen moments are the right ones
Θ_hm is a Manski identified set	Threshold, sampling-in-V, complete kernel	Same reporting logic; not-ruled-out image
Equifinal c's give wildly different θ	Width is empirical, not a theorem	They can differ; if they do not, that is a finding
A point estimate is structurally dishonest	Moral language for an estimator asked a different question	A selection from the image
The field typically reports a single number	Unscoped universal; not evidenced here	Name the tradition and what its interval contains
Cvprofiles screens out simulations	The layer screens measures, at one world	Admission of scoring rules; Θ_cv at c_0
Causal forests are SOTA for interference	They are SOTA for HTE under SUTVA	Architecture cannot see a γ edge it is not handed

Table 2. Forbidden compressions. Lectures 1–5 are written against this table. If a later sentence violates it, the sentence is wrong.

9. What this course is not

- It is not a course in building agent-based models. The kernel is one equation.
- It is not a course in do-calculus. Graphical identification is a parallel language; Lecture 4 mentions it only to put it down.
- It is not a proof of a new estimator. Aronow–Samii and Wager–Athey and Hahn–Murray–Carvalho are used off the shelf.
- It is not a literature review of equifinality. Lecture 5 names the near-neighbours so the wedge can be stated narrowly.
- It is not a forecast of the overlay. Waves have not been run.
- It is not, yet, a freeze of the specification. You will read, then ask, then we will decide what to freeze.

10. Reading order, and what each lecture is for

Lecture	Object it defines	What it refuses
1 History matching	I(c), V, τ, NROY, δ, waves	A posterior; a policy number; "observationally valid"
2 Partial identification	Identified set as a population object; Law of Decreasing Credibility	Set = confidence set; set = distribution; Θ_hm is a Manski set
3 Measurement admission	m_1, m_2, m_3, M*, Θ_cv	Screening NROY; a general proxy-validity test
4 Two foils	I_BCF, I_E2, exposure strata, the π_2 design fact	Forests as SOTA for interference; E2 on a deterministic rule

Lecture	Object it defines	What it refuses
5 Composition	The overlay; modern simulation practice; the December deliverable	A claim of uniqueness about equifinality; promised findings

Table 3. The course as five objects and five refusals.

11. Notation, collected

Symbol	Meaning
N, A, k_i	Population size 800; adjacency; degree (floored at 1)
$c = (\alpha, \beta, \gamma, \kappa, \sigma_\varepsilon)$	Free calibration; η fixed
C, NROY	Original box; not-ruled-out-yet subset after wave 3
z, M(c), I(c), τ, δ	Anchors; moment map; implausibility; threshold; discrepancy
π_0, π_1, π_2	Control; SRSWOR of 80; top-80 degree
θ_1, θ_2, τ	Random-seed contrast; hub-seed contrast; premium
$\Theta_{hm}, \Theta_{exp}, \Theta_{cv}$	Images: of θ_2 over NROY; of δ_spill over NROY; of θ_2 over M* at c_0
c_0	Pre-declared median-implausibility survivor
I_{BCF}, I_{E2}	One-world intervals: SUTVA forest / BCF; exposure-mapped Hájek
$\theta_{true}(c_0)$	Planner's number; known only because we simulate

Table 4. Notation used from Lecture 1 onward. Subscripts on Θ and I are labels, not indexes over units.

12. A last orientation before Lecture 1

If you remember four sentences from this lecture, remember these. Interference makes individual effects ill-defined; name a policy contrast. Calibration by ruling-out produces a set in parameter space, not a posterior. The honest policy report is the image of the contrast over that set, plus, separately, the image over admissible ways of scoring Y. A modern one-world interval, however honest on its own terms, answers a different question, and the project is built to say so even if, in the event, that interval covers the image.

Lecture 1 starts from the simulator's problem, not from ours: why a best-fit input is the wrong primitive once the model is not the world.

References

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